

# SAHIL KUTTE

Backend Software Engineer | Java | Spring Boot | Microservices  
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## PROFESSIONAL SUMMARY

Backend Software Engineer with 3+ years of experience building Java and Spring Boot applications, 150+ production REST APIs, microservices and event-driven systems. Experienced in service-oriented architecture, database and API performance optimization, real-time systems, external integrations, concurrency and scalable backend design. Hands-on experience building an algorithmic trading platform and healthcare-domain systems.

## TECHNICAL SKILLS

**Languages:** Java 8/17/21, SQL

**Backend:** Spring Boot, Spring Data JPA, Hibernate, REST APIs, Microservices

**Messaging / Real-time:** WebSocket, Kafka, RabbitMQ, Event-driven Architecture, Protobuf, gRPC

**Databases:** MySQL, Oracle, H2

**Cloud / DevOps:** AWS, Docker, Kubernetes, Helm, Jib, Maven

**Tools:** Git, Azure Repos, Postman, SonarQube, IntelliJ IDEA, Swagger/OpenAPI, Liquibase

**Engineering:** API Design, Database Optimization, Query Optimization, Caching, Concurrency, JWT/OAuth, System Design, Design Patterns

## PROFESSIONAL EXPERIENCE

**Backend Software Engineer / Spring Developer | Riomed Technologies India Private Limited | 10 May 2023 – Present**

- Designed and developed **150+ production-grade REST APIs** using Java, Spring Boot, Spring Data JPA and Hibernate for healthcare applications.
- Worked on microservices and service-oriented architecture, including migration of monolithic functionality toward SOA-based services and clean service boundaries.
- Implemented backend modules for medical-domain applications covering medication, patient clinical records, referrals and document management.
- Improved overall backend/API performance by **20–25%** through targeted database indexing, SQL/query optimization, reduction of unnecessary database calls and code-level optimization.
- Optimized approximately **35 database indexes** and slow-running SQL queries, reducing selected query execution times by up to **60%** (approximately **2 seconds to 800 ms**).
- Reduced redundant database round trips in paginated APIs by optimizing separate count and data-fetch operations, while implementing pagination for large datasets.
- Reduced duplicate and unnecessary processing through backend refactoring and more efficient data-access patterns.
- Designed APIs for large document uploads and retrieval, including chunked file processing and temporary-file lifecycle management.
- Integrated Microsoft Graph APIs for Teams/calendar functionality and resolved backend timezone-related issues.
- Worked with Kafka, RabbitMQ and event-driven processing while investigating concurrency, asynchronous processing and Java NIO production issues.
- Applied caching, HikariCP connection-pool configuration, database indexing and query optimization to improve backend efficiency and resource utilization.
- Used Git, Maven, Postman, SonarQube, Azure Repos and Swagger/OpenAPI throughout development, code quality and production support.

## SELECTED PROJECTS

**Algorithmic Trading Platform | Java, Spring Boot, WebSocket, Protobuf, MySQL**

- Designed an event-driven trading architecture covering real-time market data, strategy execution, risk management, order execution, positions and P&L; tracking.
- Integrated broker market data through WebSocket and Protobuf feed decoding, converting broker updates into application-level MarketTick events.
- Built an instrument master repository supporting lookup by instrument key, trading symbol, underlying and expiry for efficient option-chain operations.
- Implemented ATM option selection using underlying spot price and option-chain data, with dynamic subscription to required F&O; instruments.
- Designed thread-safe market-data caching using ConcurrentHashMap and separated market-data ingestion from strategy processing using event-driven dispatch.
- Applied **Strategy Pattern** for trading strategies, **Factory Pattern** for order creation, **Adapter Pattern** for broker integrations, **State Pattern** for order/position lifecycle handling, and **Facade-style orchestration** for subsystem interactions.
- Implemented Strategy, StrategyContext and OrderExecution abstractions to keep strategy logic independent from broker execution mechanisms.

- Built paper order execution, position management, risk validation and P&L; tracking; live broker execution and production hardening are ongoing.

### **Appointment Reminder Scheduler | Java, Spring Boot, Spring Scheduler, Cron, Batch Processing**

- Developed a scheduled backend process that automatically identifies appointments occurring one week later and triggers patient appointment reminders.
- Used Spring **@Scheduled** with cron-based execution to run the reminder job automatically, with application logic calculating the target appointment date.
- Implemented batch processing to handle multiple eligible appointments efficiently and reduce manual reminder operations.
- Designed the workflow around appointment eligibility checks, reminder generation, notification dispatch and execution logging, with duplicate-prevention and failure-handling considerations.

### **ENGINEERING HIGHLIGHTS**

- **Performance Engineering:** 20–25% overall backend/API performance improvement through indexing, query optimization, reduced DB calls and code optimization.
- **Database Engineering:** ~35 indexes optimized; selected query execution improved by up to 60% (~2s to ~800ms).
- **Concurrency:** Thread-safe collections, synchronized/volatile concepts, locks and atomic operations in concurrent processing.
- **API Engineering:** DTOs, validation, pagination, exception handling, JWT/OAuth, clean service/repository separation and API documentation.
- **System Design:** Event-driven processing, WebSocket communication, service interfaces, broker integrations and modular architecture.
- **Code Quality:** Git-based development, Maven builds, SonarQube analysis and production support.

### **CERTIFICATION**

Microservices / Backend Development Certificate — Udemy |  
[udemy.com/certificate/UC-6eedf812-6ddb-404b-b396-5f6667c91a5a/](https://udemy.com/certificate/UC-6eedf812-6ddb-404b-b396-5f6667c91a5a/)